

PHASE 3 – PROJECT DESIGN

3.1 Introduction

Project Design is the third phase of the ComicCraft AI project. In this phase, the overall architecture, workflow, frontend design, backend design, AI integration, project modules, and output flow are defined. The design is based on the requirements identified in Phase 2.

3.2 System Architecture

ComicCraft AI is organized into three primary components: Frontend, Backend, and AI Integration. The frontend collects user inputs and displays generated comics. FastAPI manages routing and processing. Gemini models generate the comic text content, while Stable Diffusion generates comic-style illustrations.

ComicCraft AI - System Architecture

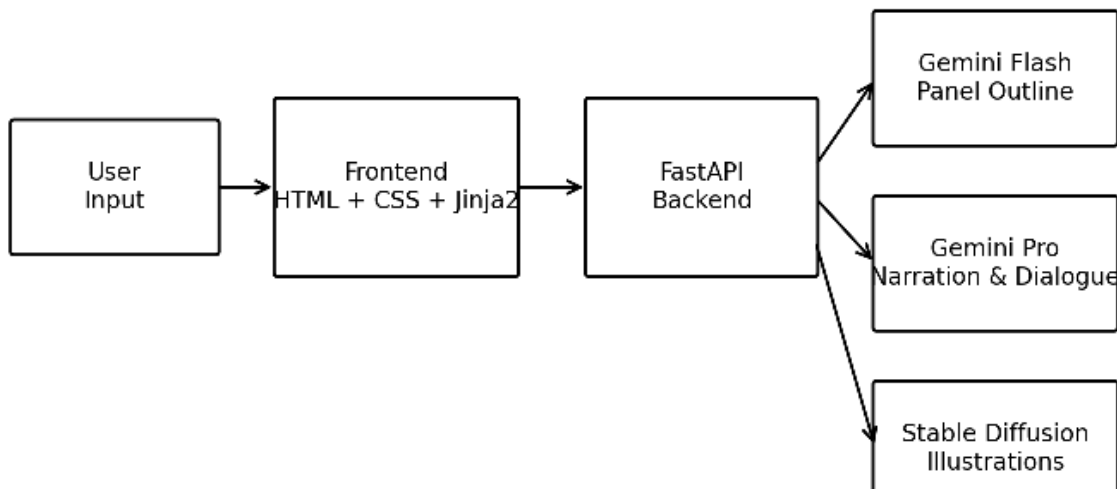


Figure 3.1: ComicCraft AI System Architecture

3.3 Frontend Design

The frontend is developed using HTML, CSS, and Jinja2 templates. The `index.html` page collects the story prompt, character name, setting, tone, and art style. The `comic_preview.html` page displays the generated panel number, title, image, scene description, captions, and narration. The `export_success.html` page provides confirmation after PDF export.

3.4 Backend Design

FastAPI is used as the backend framework. The routing logic is handled in `routes.py`. The backend receives form or JSON input, validates the request, calls the required AI modules, builds the comic layout, and exports the completed comic as a PDF.

ComicCraft AI - Frontend and Backend Design

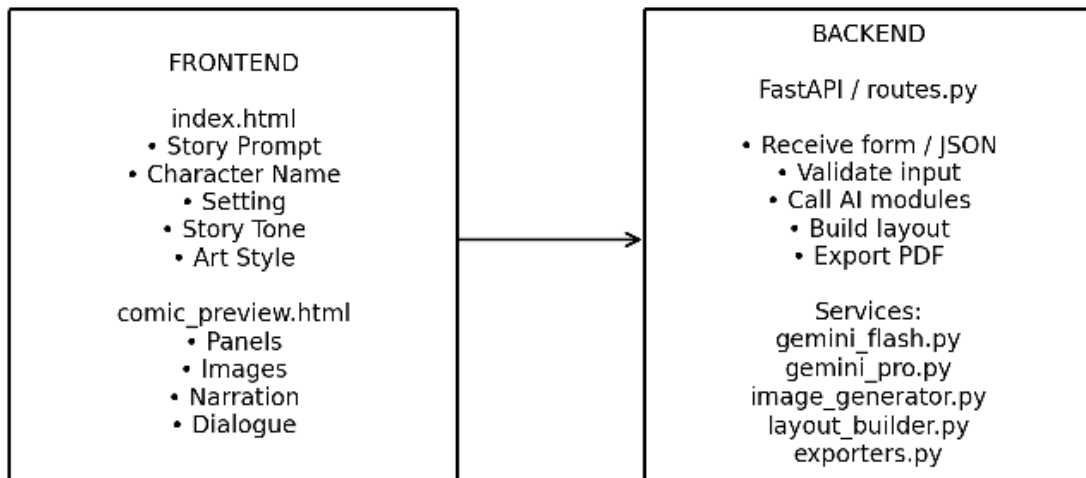


Figure 3.2: Frontend and Backend Design

3.5 AI Integration Design

The AI layer contains separate functions for different generation tasks. Gemini Flash is used to create a structured five-panel comic outline. Gemini Pro is used to expand the outline into narration and character dialogue. Stable Diffusion is used to create comic-style illustrations from image prompts.

3.6 Module Design

generate_outline() – gemini_flash.py: Generates the structured five-panel outline containing panel number, title, scene description, and image prompt.

generate_story() – gemini_pro.py: Expands the panel outlines into comic-style narration and character dialogues.

generate_image() – image_generator.py: Generates comic-style illustrations from image prompts and saves them in the panels directory.

build_comic_layout() – layout_builder.py: Matches generated images with panel stories and organizes the final comic layout.

save_pdf() – exporters.py: Compiles the comic panels, images, and narration into a multi-page PDF.

3.7 Project Workflow

The designed workflow starts with user input and continues through outline generation, story generation, image generation, layout building, preview, and PDF export.

ComicCraft AI - Project Workflow

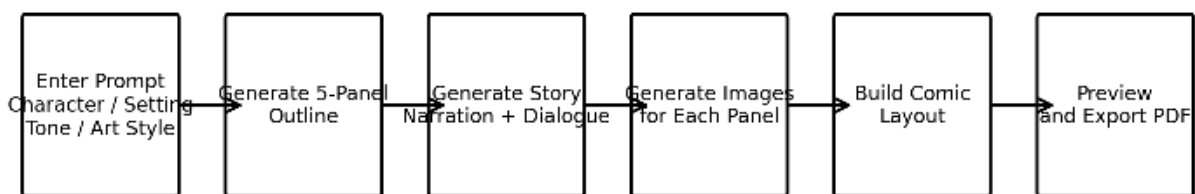


Figure 3.3: ComicCraft AI Project Workflow

3.8 Data Flow

The user submits the required comic details through the frontend. The FastAPI backend receives the data and sends the relevant information to the AI generation modules. The generated outline, narration, dialogue, and images are combined into a structured layout. The result is rendered in the comic preview and can be exported as a PDF.

3.9 Template and Route Design

The frontend templates are connected with FastAPI using Jinja2. Form elements use name attributes that map to FastAPI Form parameters, and the form submits data to the /generate route. The project also provides JSON-based comic generation and supporting routes for image testing and export confirmation.

3.10 Interface Design

The main interface is designed as a simple comic-creation form. It provides

fields for the story prompt, main character, setting, tone, and art style. After generation, the comic preview presents the generated panels with images and text. This design supports a clear flow from input to final output.

3.11 Output Design

The final output consists of a panel-by-panel comic containing generated images, narration, and dialogue. The layout builder organizes the content, and the PDF exporter produces a multi-page comic document.

3.12 Conclusion

The Project Design phase converts the requirements into a technical structure for ComicCraft AI. The architecture separates the frontend, FastAPI backend, AI services, layout generation, and PDF export into clear components. This design provides the foundation for Phase 4 – Project Planning.